The Gibbs–Thomson Effect

Study notes on Eq. 1c of Bhagat et al., *Engineering with Computers* 39 (2023) 2345–2363 — built up from the basics

> **The one-line idea.** A solid–liquid interface does not sit exactly at the melting temperature. It sits a little **colder**, because (1) a curved, sharp interface is harder to keep solid, and (2) a moving interface needs an extra push to keep atoms attaching.

1. The equation in the paper

Eq. 1c is the boundary condition that fixes the temperature on the interface Γ:

$$T_m - T_i = \Gamma_g \kappa + \frac{v_n}{\mu_k} \qquad x \in \Gamma$$

Symbol	Meaning
T_m	melting temperature (flat interface, at rest)
T_i	actual interface temperature
$T_m - T_i$	undercooling at the interface (how much colder)
Γ_g	Gibbs–Thomson coefficient = $\gamma T_m / L$ (material constant, m·K)
κ	curvature of the interface (≈ 1/radius; 0 for flat)
v_n	normal velocity of the interface
μ_k	interface-attachment (kinetic) coefficient

Strictly, only the first term, $\Gamma_g \kappa$, is the **Gibbs–Thomson effect**. The second term, v_n/μ_k, is **kinetic undercooling**. They are usually written together because both lower the interface temperature. Eq. 1b (the Stefan condition) says how fast the interface moves; Eq. 1c says what temperature it sits at while moving.

2. Building blocks

Melting temperature

At T_m (0 °C for ice) solid and liquid can sit side by side with neither taking over. This is true only for a **flat** interface that is **not moving**.

Undercooling

Liquid can be cooled below T_m without freezing (supercooled water). The gap $\Delta T = T_m - T$ is the **undercooling**. The bigger it is, the stronger the urge to freeze — it is the driving force for solidification.

Interface

The boundary between solid and liquid. Its temperature is T_i. Eq. 1c tells us $T_i < T_m$ and by how much.

3. Term 1 — curvature: $\Gamma_g \kappa$

Intuition: sharp tips have a weak grip

Each atom in a solid is held by bonds to its solid neighbours. An atom on a flat surface has many solid neighbours; an atom on a sharp tip has few, with liquid all around. A weakly held atom falls off (melts) easily — which is why the corners of an ice cube round off first.

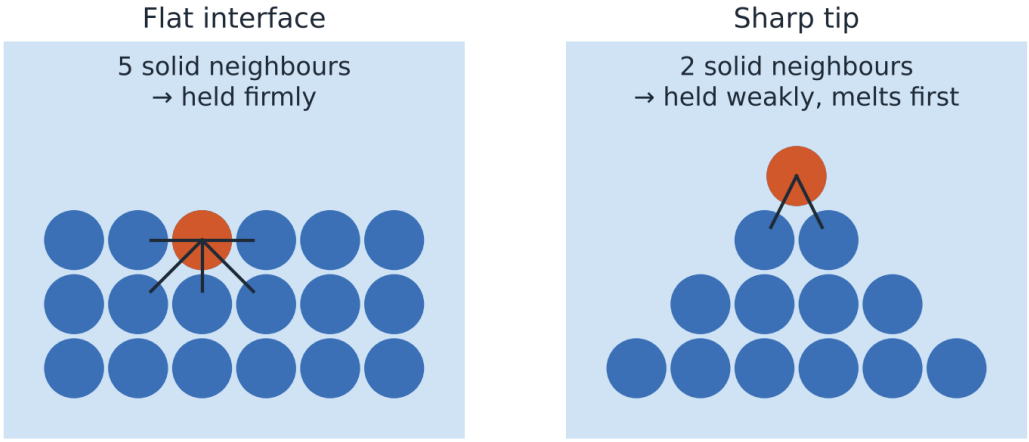


Figure 1. Fewer solid neighbours at a sharp tip means a weaker grip.

Think of coldness as a *reward* for staying solid. A flat-surface atom is satisfied with the small reward available just below T_m; a tip atom needs a bigger reward, i.e. a colder temperature. **Sharper tip → weaker grip → must be colder.**

Derivation: a tug-of-war between reward and penalty

Step 1 — the reward (bulk driving force). The free-energy difference per unit volume between liquid and solid is $\Delta G_v = \Delta H - T\Delta S$. At T_m the phases coexist, so $\Delta G_v = 0$, giving $\Delta S = L/T_m$ (L = latent heat per volume):

$$\Delta G_v = \Delta H - T\Delta S, \qquad \Delta G_v(T_m) = 0 \Rightarrow \Delta S = \frac{L}{T_m}$$

Below T_m:

$$\Delta G_v = L - T\frac{L}{T_m} = \frac{L(T_m - T)}{T_m} = \frac{L\Delta T}{T_m}$$

So the reward for freezing is simply **proportional to how cold it is**.

Step 2 — the penalty (surface energy). Creating interface costs γ per unit area (surface tension). What matters is how much new surface you make for each bit of new solid. Let a circular tip of radius r grow by dr:

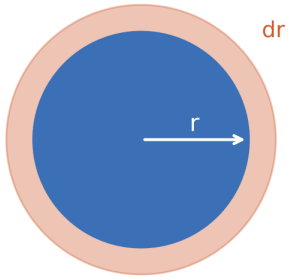


Figure 2. Growing a circle of radius r by dr.

$$\frac{\text{surface energy added}}{\text{solid added}} = \frac{\gamma \cdot 2\pi\, dr}{2\pi r\, dr} = \frac{\gamma}{r} = \gamma\kappa \qquad (\kappa = 1/r)$$

In 3D (a sphere) the same idea gives:

$$\frac{\gamma \cdot 8\pi r\, dr}{4\pi r^2\, dr} = \frac{2\gamma}{r} = \gamma\kappa \qquad (\kappa = 1/r_1 + 1/r_2 = 2/r)$$

This is the “weak grip” in mathematical form: a small r means the tip is mostly surface, so each bit of solid added carries a big surface penalty.

Step 3 — balance. The tip is in equilibrium when reward = penalty:

$$\frac{L\,\Delta T_\Gamma}{T_m} = \gamma\kappa \quad\Longrightarrow\quad \Delta T_\Gamma = \frac{\gamma T_m}{L}\kappa = \Gamma_g \kappa$$

Result: $\Delta T_\Gamma = \Gamma_g \kappa$ with $\Gamma_g = \gamma T_m / L$. A sharper interface (larger κ) needs more undercooling to stay solid.

How big is it?

Using $\Gamma_g = 2.36 \times 10^{-7}$ m·K from the paper’s alloy parameter table and a tip radius of about 1 µm:

$$\Delta T_\Gamma = \Gamma_g \kappa \approx (2.36 \times 10^{-7}\,\text{m K}) \times \frac{1}{10^{-6}\,\text{m}} \approx 0.24\ \text{K}$$

Only a fraction of a kelvin — yet it is what selects the tip radius and stabilises the dendrite shape. Without it, the interface would keep sharpening without limit.

Sign convention

- Convex solid bulging into the liquid (a dendrite tip): $\kappa > 0 \rightarrow$ interface is colder than T_m.
- Concave regions (the grooves between dendrite arms): $\kappa < 0 \rightarrow$ interface can be slightly warmer than T_m.
- Flat interface: $\kappa = 0 \rightarrow$ no curvature correction.

4. Term 2 — kinetics: v_n/μ_k

Intuition: the doorway

At the interface atoms constantly jump both ways: liquid → solid (joining) and solid → liquid (leaving). Like people passing through a doorway, the room only fills if more go in than come out.

- **At T_m:** joining = leaving → the interface does not move.
- **Slightly below T_m:** joining slightly beats leaving → slow growth.
- **Well below T_m:** joining strongly beats leaving → fast growth.

Coldness creates the imbalance, and the imbalance sets the speed. **Faster growth → bigger imbalance needed → must be colder.**

Derivation

Step 1. Joining happens at a rate $\propto e^{-Q/k_BT}$; leaving at a rate $\propto e^{-(Q+\Delta g)/k_BT}$, where Q is an activation barrier and Δg is the free energy gained per atom on freezing ($\propto$ local undercooling ΔT_μ). The net speed is the difference:

$$v_n = v_0[\,1 - e^{-\Delta g/k_BT}\,], \qquad \Delta g \propto \Delta T_\mu$$

Step 2. For small undercooling, linearise the exponential:

$$e^{-x} \approx 1 - x \implies v_n \approx v_0 \frac{\Delta g}{k_B T} = \mu_k \Delta T_\mu \implies \Delta T_\mu = \frac{v_n}{\mu_k}$$

Unlike the curvature term, this is a **constitutive assumption** (linear kinetics), valid for small undercooling. μ_k is the slope: how easily atoms attach.

> **Result:** $\Delta T_\mu = v_n/\mu_k$. In metals μ_k is very large (atomically rough interfaces accept atoms easily), so this term is usually tiny — which is why the paper neglects attachment kinetics in its alloy model.

5. Putting the two together

Treat the interface undercooling as a budget split between two jobs. Both effects are small and independent, so to first order they simply add — giving Eq. 1c.

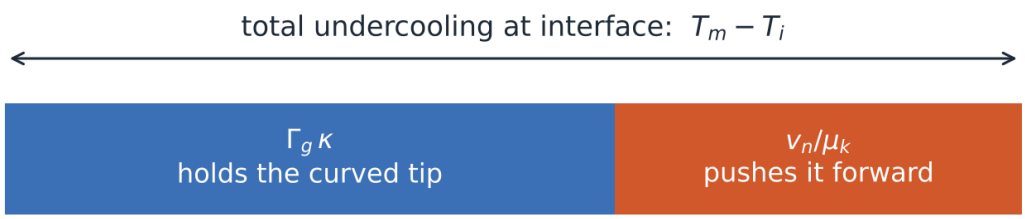


Figure 3. The undercooling budget at the interface (proportions illustrative).

$$T_m - T_i = \Gamma_g \kappa + \frac{v_n}{\mu_k} \qquad x \in \Gamma$$

A dendrite tip is both sharp and moving, so both terms act on it. Together with heat flow (Eqs. 1a–1b) this condition decides how pointy the tip becomes and how fast it grows.

6. Where it appears elsewhere in the paper

Non-dimensional form — Eq. 2c

Using the scaled temperature $u = c_p(T - T_m)/L$, Eq. 1c becomes:

$$u^* = -d(\mathbf{n})\kappa - \beta(\mathbf{n}) v_n, \qquad d(\mathbf{n}) = \frac{\gamma(\mathbf{n}) T_m c_p}{L^2}$$

- $d(\mathbf{n})$ is the **capillary length** — the non-dimensional version of Γ_g.
- $\beta(\mathbf{n})$ is the **kinetic coefficient** — the non-dimensional version of $1/\mu_k$.
- The minus signs appear because u is negative when the interface is below T_m.

Anisotropy: why dendrites have preferred directions

Surface tension depends on crystal orientation, $\gamma(\mathbf{n})$, so the Gibbs–Thomson correction does too. The paper uses four-fold anisotropy:

$$a(\mathbf{n}) = 1 + \varepsilon_4 \cos(4\theta), \qquad \varepsilon_4 = 0.05$$

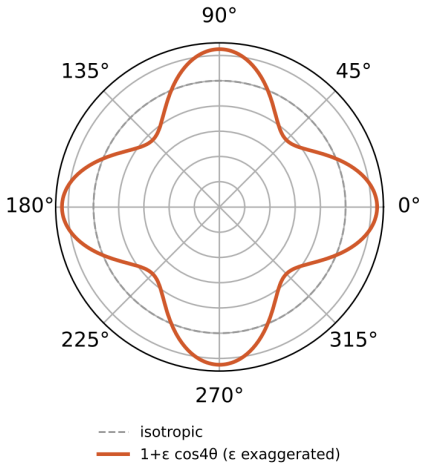


Figure 4. Four-fold anisotropy (strength exaggerated for visibility). Growth is favoured along the lobes, giving four main dendrite arms.

Without anisotropy, curvature alone would not pick a growth direction and the interface would form irregular, seaweed-like shapes. With it, dendrites grow along preferred crystal directions.

Alloy solidification — Eqs. 7c and 8c

For the binary alloy, the interface temperature is set by curvature (Gibbs–Thomson), cooling rate and the imposed temperature gradient; attachment kinetics are dropped. Written in terms of supersaturation u_c:

$$u_c^* = -d_0\kappa + \dot{\theta}t - \gamma z, \qquad d_0 = \frac{\Gamma_g}{\Delta T_0}$$

Here $d_0 = \Gamma_g/\Delta T_0$ is the **chemical capillary length** — the same Gibbs–Thomson idea, now scaled by the alloy's freezing range ΔT_0.

7. Quick-recall summary

	Curvature (Gibbs–Thomson)	Kinetic undercooling
Term	$\Gamma_g \kappa$	v_n / μ_k
Physical cause	Surface energy of a curved interface	Net attachment of atoms needs a driving force
Analogy	Sharp tip → weak grip	Doorway: more in than out
Origin	Energy balance (thermodynamics) — derived	Linearised rate law — assumed for small ΔT
Grows with	Sharpness (κ)	Speed (v_n)
Non-dim. (Eq. 2c)	$d(\mathbf{n})\,\kappa$	$\beta(\mathbf{n})\,v_n$
In the alloy model	Kept ($d_0\,\kappa$)	Neglected (μ_k large in metals)

> **Remember:** $T_m - T_i$ = (extra cold to hold a sharp interface) + (extra cold to move it). Sharper → colder. Faster → colder.